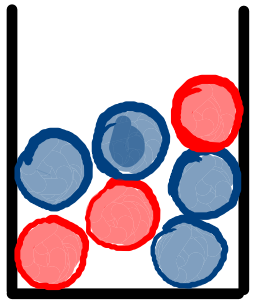

7 bolillas → 4 azules
→ 3 rojas

<u>Experimento 1</u>: se toma 1 bolilla al azar

$$X = \begin{cases} 1 & \text{sale roja} \\ 0 & \text{sale azul} \end{cases} \qquad R_X = \{0, 1\}$$

regla de Laplace

$$p_X(0) = P(X=0) = \frac{4}{7} = 1-p = q$$

$$p_X(1) = P(X=1) = \frac{3}{7} = p$$

$$\mu_X = E[X] = \sum_{x \in R_X} x\, p_X(x) = 0 \cdot q + 1 \cdot p = p$$

$$E[X^2] = \sum_{x \in R_X} x^2\, p_X(x) = 0^2\, q + 1^2 \cdot p = p$$

$$\sigma_X^2 = Var(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1-p) = pq$$

V.A. Bernoulli:

$X \sim \text{Bernoulli}(p) \rightarrow R_X = \{0,1\}$

"tiene distribución"
"está distribuido como"

$P_X(0) = q = 1-p$

$P_X(1) = p$

$\mu_X = p$

$\sigma_X^2 = pq$

Modelar experimentos con una probabilidad de éxito p

↳ (sacar una bola roja)